

TEJAS AI

The autonomous brain for cooling & energy

तेजस् · Built in India · On-prem · Savings-share

What did your building know last night that you didn't?

Your BMS knew the chiller was short-cycling.

Your historian knew inlet temps were drifting.

Your weather feed knew Thursday would be 47°C.

The data was there. The story wasn't.

Cooling is the invisible crisis

30–40% of facility power · controlled by 1990s logic

- Data centers, cold chain, hospitals, factories — largest energy line item is removing heat
- Rule-based BMS: if temp > X, turn on chiller — reacts after heat builds, over-cools 'to be safe'
- Operators are not data scientists — they can't tune controllers or read 8,000-point historians
- Every predictive signal exists in the building — nobody connects it into one narrative

30–40%

of DC power = cooling

20–28%

energy wasted vs optimal

14 mo

GPU-16 drift before trip

Why India. Why now.

The perfect storm — and the science is proven

- 40–48°C summers · expensive grid · water scarcity · DC + cold-chain explosion
- 2024 Indian heat wave — longest on record; engineers warned of grid blackouts as AC load spiked
- DeepMind cut Google cooling ~40%; Phaidra runs NVIDIA AI factories — category validated
- Phaidra chases hyperscalers — thousands of cold stores, edge DCs, factories have no one optimising

Real anchor: Chennai DC corridor · CtrlS Tier-IV · OMR/Siruseri belt

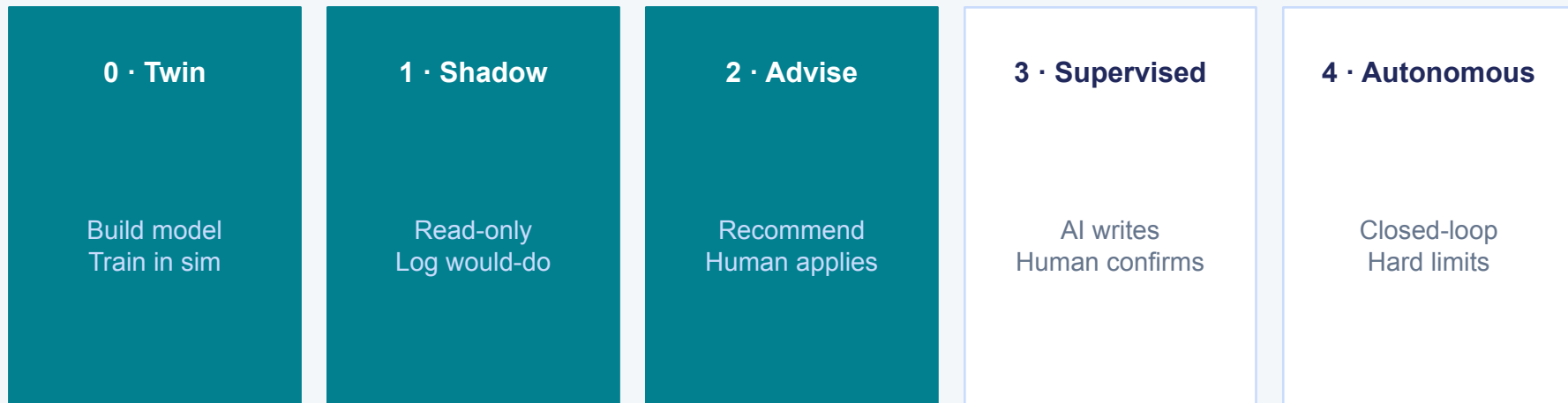
Tejas AI

Predict · optimise · explain · act — on a digital twin first

- Self-learning agent replaces dumb thermostat logic — feed-forward, not reactive
- Calibrated digital twin: topology of racks, chillers, CDUs, cold rooms — tuned to historian ($\pm 1^{\circ}\text{C}$)
- Train and prove in simulation before touching hardware — zero risk to start
- Operator talks in English, Hindi, Tamil — dashboard builds from voice, no training
- Predictive maintenance + Field AR — draft work order, guide technician to exact part

Deploy safely — the trust ladder

Conservative operators never let unproven AI near critical cooling



Safety Supervisor · rate limits · sensor dropout → instant revert to baseline · every action logged

One platform — every facility that removes heat

Digital twin + AI brain — running today

Data Center

Autonomous cooling
PUE · GPU-16 PM

Smart Factory

PRESS-7 fault
Phone scan twin

Geothermal

SEP-2 separator
Reinjection loop

Cold Chain

2–8°C band hold
Fleet projection

+ Field AR · VR walkthrough · 6-step brownfield ingestion · Twin Studio

See it live — one laptop, offline

`cd tejas-twin && ./run.sh → localhost:7878`

- 3D hall: racks glow by temperature · sky = weather · airflow particles
- Drag to 47°C: baseline panics · Tejas holds band · PUE 1.3 vs 1.5
- Ask: 'Is any machine going down?' → GPU-16 · parts · service email
- Ask: 'Project savings across 50 sites' → fleet panel
- Field AR on phone · VR first-person walk · Hindi voice

DEMO NOW — 90 seconds — then return to deck

Stories grounded in real incidents

Demo characters are composites — inspired by these

Chennai DC

Heat wave + chiller short-cycle
CtrlS corridor · 2024 grid stress

The Geysers, CA

Reinjection after steam depletion
Separator SEP-2 stand-in

Renesas Naka 2021

Fab fire · 30% auto MCU supply
100+ day restart

India cold chain

₹92,651 Cr annual post-harvest loss
Refrigeration = 45% of bill

The numbers that close deals

12 MW hall · India tariff · validated in shadow pilot on your data

**₹1.8–
3.2Cr**

cooling saved / year

₹2+ Cr

one avoided outage

₹8.3L

one PM fix / year

₹90–160Cr

50-site fleet / year

- PUE at 47°C: ~1.50 baseline → ~1.30 Tejas (–13%) — same ASHRAE safety band
- Cold storage: 15–25% refrigeration optimisation → ₹1.4–2.4 Cr/year on ₹65L/month bill
- Water: lower WUE in Chennai-class climates — licence to operate
- Carbon: hundreds of tonnes/hall/year — auditable for ESG reporting

Business model

Aligned incentives · land-and-expand

Savings-share

20–40% of verified savings
Primary · zero capex objection

SaaS / facility

Recurring once trust built
Predictable for chains

Twin-build

Paid on-ramp · calibration
Produces ROI case

One hall → whole campus → fleet. NRR engine.

Why we win

Not the dashboard — the calibrated twin + safe control + site data

	Status quo BMS	Phaidra / foreign	Tejas AI
Approach	Reactive rules	Hyperscaler cloud	Predict + on-prem
Market	Everyone	Top 50 DCs	Mid-market India + world
Deploy	Rip-and-replace fear	Premium SaaS	Shadow → control
Moat	None	Scale + capital	Twin + policy + data

- Sovereign: Ollama on-prem — data never leaves site
- India-tuned: vernacular voice, Indian .epw weather, operator workflows

Go-to-market

Proof and spectacle — not cold sales

- 1. Clip the demo — heat-wave moment spreads online (30–60 sec)
- 2. Shadow pilots — 30-day read-only on customer historian; undeniable ROI
- 3. Savings-share — procurement killer; pay from savings only
- 4. Partnerships — cold-chain assoc., DC operators, ESCOs, BEE programs
- 5. Content — open benchmark; 'What your BMS knew last night' series

The ask

[Customise: round size · use of funds · milestones]

- Raising: ₹[X] [pre-seed / seed]
- 40% pilot deployments + calibration engineering
- 30% twin engine + control policy hardening
- 20% GTM — demo tour, content, partnerships
- 12-month goal: 5 shadow pilots · 2 paid control contracts

We don't sell a dashboard.

**We sell rupees off the power bill —
for everyone the giants ignore.**